APPROXIMATING THE EQUILIBRIUM EFFECTS OF INFORMED SCHOOL CHOICE: ONLINE APPENDIX

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O-1. *Provision of Preschool Services in Chile*

Chile's preschool sector combines public and private providers, with enrollment rates that have risen substantially but remain below OECD averages.[1] Two public networks, Junji (approximately 3,000 centers) and Fundación Integra (approximately 1,000 centers), provide tuition-free services and are explicitly tasked with reaching children from low-income families nationwide. In 2016, 42% of enrolled 3–4-year-olds attended Junji centers and 18% attended Integra centers; the remaining children attended private providers, some publicly funded and others charging tuition. Admission to both public and private centers is decentralized, with public providers giving priority to lower-income families (Aguirre, 2011). Several targeted public programs also support families in sending children to preschool. Until 2016, Junji also carried out light monitoring of private providers, primarily related to inputs; an independent quality-assurance agency has since assumed that role. Expenditure per student was $6,408 in 2015, of which 15% came from private sources (OECD, 2016).

O-1.1. *Fundación Integra*

Fundación Integra (hereafter Integra) is the second-largest public preschool network in Chile, serving more than 72,000 children aged three months to four years in tuition-free centers con-

[1]This subsection draws on OECD (2016). As of 2015, enrollment of 0–2 year-olds in formal care was 18% (vs. 33% across the OECD), rising to 54% for 3-year-olds (vs. 71%), nearly double the 2005 rate but still below the OECD average of 71%.

centrated in low-income neighborhoods. Integra does not offer primary education, so its oldest cohort must choose a primary school each year. This feature makes Integra an ideal partner for studying school choice: we can reach families precisely when they are deciding where to enroll, using the network's existing infrastructure for data collection and intervention delivery, without the selection concerns that would arise from recruiting at primary schools themselves.

O-2. *Sample Selection*

We worked in Chile's three largest regions (Valparaíso, Biobío, and Santiago), selecting urban Integra preschools located in areas with meaningful school competition. Specifically, all preschools in municipality i were eligible if there were at least 10 primary schools within a 2 km radius and the ratio $(\text{primary schools}_i/\text{preschools}_i) \geq 2$. These criteria were operationally important because the intervention gave each family a personalized report card comparing nearby schools: in places with few local alternatives, the report card could not present a meaningful choice set. We further restricted attention to schools in the three lowest SES categories (as classified by the Ministry of Education), which constitute nearly all schools in the effective choice set of families in the first three income quintiles.

These criteria yielded 143 eligible preschools. We randomly assigned them to treatment and control groups, stratifying by region, grade level, and the number of primary schools within 2 km. We then contacted each preschool to schedule a visit during an existing parent meeting between August and December 2010.

The archived fieldwork file identifies 41 municipalities represented by these eligible centers but does not retain the complete pre-screening list needed to reconstruct an exact 2010 denominator. For scale, we therefore use the closest available broader Integra inventory as a benchmark. The official 2018 preschool census contains 273 Integra centers in the same 41 municipalities, of which 257 are urban. The 143-center experimental frame is 56% of that later urban count. This is a descriptive benchmark rather than a count of centers that failed the 2010 screen: the census is eight years later, and centers may have opened, closed, moved, or changed identifiers in the interim.

Of the 143 preschools, 10 could not be visited, owing to principal refusal, scheduling conflicts, or low parent attendance. Table O-1 compares observable characteristics of preschools with and without meetings. The 10 excluded centers have a somewhat higher share of mothers with tertiary education, a higher share of families in the second income quintile, and a lower

share of indigent families. Because no parents at these centers signed the informed consent, the effective sample consists of 133 preschools.

TABLE O-1

DIFFERENCES BETWEEN PRESCHOOLS WITH AND WITHOUT MEETINGS

	With Meeting	Without Meeting	Difference	t-stat
Enrollment	40.21	46.50	-6.29	-1.02
Mean Attendance	27.99	32.40	-4.41	-0.77
Mother w. Complete Tertiary (%)	9.07	12.81	-3.74	-1.76*
Mother w. Complete Secondary (%)	47.74	45.36	2.38	0.70
Mother w. Incomplete Secondary (%)	34.72	34.89	-0.17	-0.05
Mother w. Complete Primary (%)	7.81	6.94	0.87	0.65
Q1 of Income (%)	58.35	53.89	4.47	1.22
Q2 of Income (%)	31.56	38.42	-6.86	-2.02**
Q3 of Income (%)	8.00	5.95	2.05	1.27
Indigent (%)	15.37	9.68	5.69	2.73***
Poor non-indigent (%)	40.67	42.10	-1.42	-0.41
Non-poor (%)	43.96	48.23	-4.27	-1.33
Number of Preschools	133	10	143	

Notes: Columns 1–2 report averages for preschools with and without meetings. Column 3 reports the difference; Column 4 reports the t-statistic for the null of equal means, with standard errors clustered at the preschool level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In the 133 participating preschools, 1,832 parents signed informed consent and completed a baseline survey before any information was distributed. The survey asked whether the family planned to enroll the child in primary school in 2011, whether a school had already been chosen, whether enrollment was complete, and whether older siblings already attended primary school. The enrollment split used in the analysis is drawn from the follow-up survey's retrospective question about status at the meeting, rather than treating the baseline item as the analysis variable. Because visits were not announced in advance, meeting attendance was plausibly independent of treatment status. Table O-2 shows that attendance at parent meetings was similar across treatment and control preschools, with nearly all parents present signing the informed consent (only 27 of 1,859 declined, or 1.4%).

TABLE O-2

ATTENDANCE TO PRESCHOOL AND PARENT MEETING

	Control	Treatment	Total
Enrollment (year)	41.47	39.57	40.21
Mean attendance (year)	28.69	27.64	27.99
Attendance to meeting	13.24	14.35	13.98
Number of preschools	45	88	133

Notes: Columns 1–2 report averages for control and treatment preschools; column 3 reports the overall average. Mean attendance to the meeting represents roughly 35% of enrolled students and about half of typical daily attendance. Visits were not announced in advance.

O-2.1. *Older Siblings and School-Entry Timing*

Older siblings may constrain school choice. We link experimental children to other children of the same mother in administrative enrollment records and compare the resulting measure across the enrollment-timing groups and with the population used in the demand model. Because many Chilean primary schools also offer pre-kindergarten or kindergarten, we define school entry as the first year the child appears at the school ultimately attended in first grade. Among experimental children for whom this entry year is observed, 70.6% first appear at that school in pre-kindergarten or kindergarten.

O-3. *Treatment Design and Implementation*

The intervention had two components: a personalized report card and a short video. This subsection describes each in detail.

Report card. Each treated family received a report card listing characteristics of nearby primary schools.[2] The card reported: (i) test scores, averaged over Math and Reading for 2006–2009 to reduce noise from any single year; (ii) the trend in test scores across years, since a school at the median that has substantially improved may be a better match for some parents than one with the same level score that has worsened; (iii) the official tuition charged to parents, from

[2]The report card included up to 30 schools in the lowest three SES categories within 2 km of the preschool. Higher-SES schools were excluded because they are generally not in the effective choice set of families in this context. When more than 30 schools fell within 2 km, we randomly dropped schools from the interior of the quality distribution, retaining the highest- and lowest-scoring schools so that the card preserved the full range of local school quality.

TABLE O-3

OLDER SIBLINGS, ENROLLMENT TIMING, AND THE MODEL SAMPLE

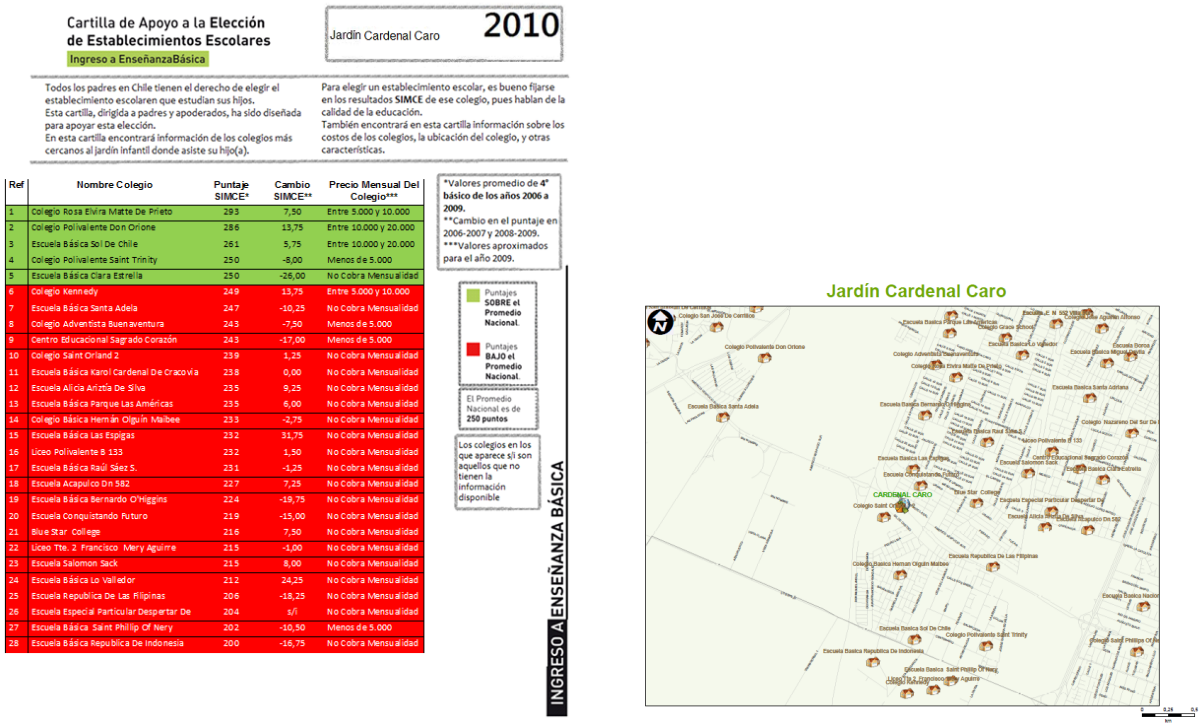
	Near intervention		First appearance at eventual school		
Sample	Linked N	Older sibling	Linked N	Older sibling	Same school
All experimental children	1,528	0.340	1,484	0.336	0.719
Already enrolled	509	0.346	500	0.334	0.749
Not yet enrolled	829	0.333	804	0.333	0.713
Structural estimation sample	705	0.321	684	0.319	0.725
Population: 53 markets, types 1–4	—	—	203,366	0.371	0.756
Selection and robustness estimates					
Older sibling → already enrolled					-0.009 (0.026)
Older sibling → structural-sample inclusion					-0.032 (0.028)
Treatment × older sibling → chosen-school quality					0.038 (0.061)
Quality ITT: baseline specification					0.0930 (0.0365)
Quality ITT: adding sibling status					0.0936 (0.0369)

Notes: An older sibling is another child with the same masked mother identifier, an earlier birth year, and enrollment in grades 1–8. Administrative sibling rates condition on a valid mother link. "Near intervention" uses 2011 enrollment, the administrative year closest to the preschool meetings. "First appearance" is the earliest year in 2011–2013 that the child is observed at the school ultimately attended in first grade and includes pre-kindergarten and kindergarten. The population row contains first-grade entrants of model types 1–4 in the 53 demand-model markets and 2012–2013 cohorts. "Same school" is the share of children with an observed older sibling who enter that sibling's school; it is an outcome, not a predetermined control. Selection and heterogeneity regressions include the paper's controls and municipality fixed effects, with standard errors clustered by preschool. Structural-sample inclusion is estimated among not-yet-enrolled families. The quality outcome is the normalized value-added measure used in the model.

Ministry of Education records;[3] (iv) school type (public or private); and (v) school location. A map showing all listed schools accompanied each card. To signal relative quality, schools above the national mean test score (roughly 250 points) were colored green and those below were colored red. This design feature reflected policymaker preferences for the planned intervention and was intended to address the potential asymmetry of information parents face regarding school quality. Figure O-1 shows an example.

[3]Official tuition does not capture all costs families face (e.g., materials fees, parent-association dues) or discounts that individual schools may offer. We used it because it is an objective, comparable measure available for all schools.

FIGURE O-1.—Report Card and School Map



Notes: Left panel: front of the report card, listing each school's average test score, the change in its score over time, and its monthly tuition band; the back of the card (not shown) lists each school's address, grades offered, and sector. Right panel: accompanying map with the location of nearby schools. Green: schools above the national mean test score; red: below.

Video. The video featured three role models from low-income backgrounds. The first was a mother who had decided to transfer her second-grade son to a school with higher test scores in order to give him a better education. The second was a college student completing a degree in computer engineering who had attended a strong high school in a low-income neighborhood. The third was a young woman from a similarly disadvantaged background who, partly because of her high school, had completed vocational training and now held a stable job at a bank. All three were chosen to convey that good schools and higher education are accessible to low-income families, consistent with the evidence in Nguyen (2008) on the effectiveness of role models from similar backgrounds. The video also presented descriptive evidence on the returns to tertiary education in Chile (for example, that college graduates earn roughly three times the average earnings of those with only a high school diploma) and highlighted the observed correlation between primary school quality and later enrollment in higher education, without claiming a causal relationship (Jensen, 2010).

FIGURE O-2.—Video: Role models from low-income neighborhoods



(a) Parent (b) College student (c) Young professional

Notes: Screenshots from the video shown to treated families. Panel (a): a mother who transferred her son to a higher-performing school. Panel (b): a computer engineering student from a low-income neighborhood. Panel (c): a young woman who works as a commercial assistant. All three role models come from low-income backgrounds.

FIGURE O-3.—Video: Importance of school choice for future opportunities



Notes: These panels summarize the part of the intervention that emphasized the long-run returns to school choice. The video encouraged parents to think about their child's future education and employment opportunities and linked school quality to preparation for higher education and better jobs.

The video thus complemented the report card's factual content with motivational framing: even families who know which schools score well may underweight quality in their decisions if they do not perceive the returns to be large or attainable.

O-3.1. *Direct Effects of Report Card Design*

TABLE O-4

DIRECT EFFECTS OF REPORT CARD DESIGN

	On Report Card		Green		Top 5	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Full Sample						
Treatment	-0.118***	-0.112***	0.039	0.028	0.075*	0.064*
	(0.029)	(0.030)	(0.046)	(0.044)	(0.041)	(0.039)
N obs.	1775	1775	1136	1136	1136	1136
Panel B: Enrolled sample						
Treatment	-0.007	0.004	0.069	0.038	0.048	0.025
	(0.054)	(0.053)	(0.076)	(0.072)	(0.069)	(0.069)
N obs.	596	596	389	389	389	389
Panel C: Not enrolled sample						
Treatment	-0.172***	-0.159***	0.019	0.036	0.076*	0.090**
	(0.032)	(0.033)	(0.059)	(0.060)	(0.041)	(0.041)
N obs.	975	975	639	639	639	639
Randomization controls	×		×		×	
Expanded controls		×		×		×

Note: Randomization controls include market characteristics of schools (number and test scores mean, standard deviation and percentiles 25, 50 and 75.). Expanded controls include Mother's education, household information (size, durable goods, owned house), baseline school choice information.

Notes: ITT estimates from regressions of report-card-specific outcomes on treatment assignment. Standard errors in parentheses are clustered at the preschool-center level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table O-4 asks whether specific design features of the report card affected school choice beyond the information content of the intervention. Columns 1–2 test whether treated families became more likely to choose a school that was listed on the card. Columns 3–6 are estimated among families who chose a school that appeared on the report card, since the green and top-five classifications are only defined for listed schools. Columns 3–4 test whether, conditional on choosing a listed school, treated families were more likely to choose one coded in green (above the national mean test score). Columns 5–6 test whether they were more likely to choose one of the top-five schools highlighted on the card. Within each pair, the first column includes the randomization controls and the second adds the expanded family controls.

The results provide little evidence that these design features mechanically drove choices. If anything, treated families are less likely to choose a school that appears on the card, especially in the not-yet-enrolled sample, which is consistent with the main-text finding that treated fami-

lies travel farther in order to attend better schools. The estimates on choosing a green school are small and statistically insignificant. By contrast, the estimates in columns 5–6 suggest that treated families are somewhat more likely to choose a school ranked among the top five in the card, particularly among those who had not yet enrolled. This conditional association does not separately identify a causal effect of highlighting: school rank and visual presentation were not independently randomized, and choosing a listed school is itself affected by treatment.

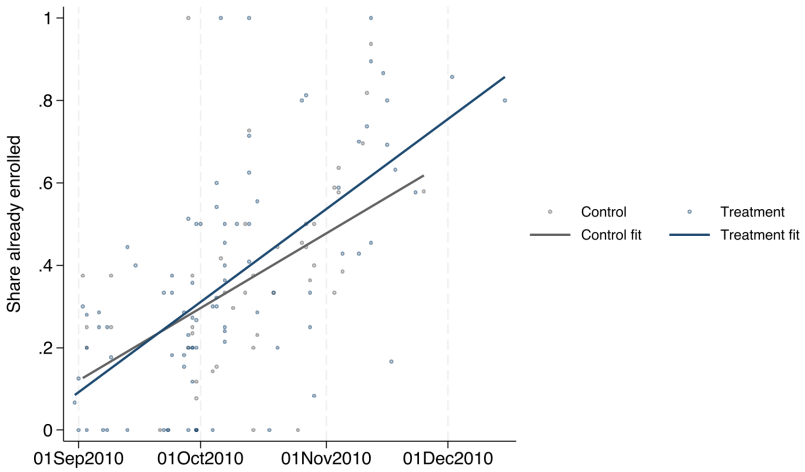
O-4. *Selection on Reported Enrollment Status*

A substantial share of families reported in the follow-up survey that they had already enrolled their child in a primary school by the time of the preschool meeting. Because the reduced-form analysis distinguishes between already-enrolled and not-yet-enrolled families, and because the structural estimation uses only the latter, it is important to examine the relationship between this classification, meeting timing, and observed family characteristics.

The meetings were conducted over a 16-week window between August and December 2010. Figure O-4 shows that the share of families reporting prior enrollment rises sharply with the calendar date of the meeting: roughly 20% in September and 65% by November. The pattern is similar across treatment and control groups, indicating that the ordinary progression of the enrollment season is an important determinant of the split. It does not establish that the two subgroups are otherwise exchangeable.

Table O-6 compares observables across enrolled and not-yet-enrolled families. The two groups are balanced on mother's education, poverty status, household size, and birth outcomes. Enrolled families report somewhat more durable goods and are marginally more likely to have had a hospital birth, but the differences are small.

FIGURE O-4.—Timing and Probability of Matriculation



O-5. *Balance Tables and Summary Statistics*

TABLE O-5

TREATMENT BALANCE AT THE SCHOOL LEVEL

	Difference T-C		Control Group Mean	
	(1)		(2)	
Enrollment	-1.898	(3.248)	41.467	(2.530)
Mean attendance	-1.053	(2.430)	28.689	(1.903)
Mother HE	-0.643	(1.552)	9.495	(1.320)
Mother HS	-0.915	(2.195)	48.347	(1.652)
Mother NHS	0.760	(1.010)	7.309	(0.697)
Q1 Income	0.577	(2.996)	57.970	(2.348)
Q2 Income	0.288	(2.142)	31.365	(1.587)
Q3 Income	-1.136	(1.216)	8.752	(0.930)
Very Poor	0.637	(1.865)	14.947	(1.406)
Poor	0.083	(2.233)	40.619	(1.816)

Notes: Column 1 presents the coefficient and standard error for the difference between the treatment and control groups in a regression of each variable on treatment status. Column 2 presents the coefficient and standard error for the control group mean. * p-value<10% ** p-value<5% *** p-value<1%.

TABLE O-6

CORRELATES OF REPORTED ENROLLMENT STATUS AT THE PRESCHOOL MEETING

	Difference Enrolled - Non-Enrolled		Non-Enrolled Group Mean	
	(1)		(2)	
Panel A: SES characteristics				
Household size	-0.035	(0.108)	4.927	(0.081)
Durable goods	0.382***	(0.116)	4.461	(0.080)
Owns Dwelling	0.047	(0.031)	0.343	(0.019)
Mother head of hh	0.001	(0.030)	0.832	(0.014)
Mother NHS	-0.010	(0.022)	0.192	(0.016)
Mother HS	-0.039	(0.025)	0.391	(0.019)
Mother HE	0.007	(0.019)	0.836	(0.016)
Poor	-0.013	(0.016)	0.895	(0.011)
Another child in primary	0.010	(0.029)	0.405	(0.018)
Panel B: Birth characteristics				
Gestation Weeks	-0.019	(0.094)	38.751	(0.056)
Birth Weight	-3.982	(25.338)	3,342.137	(15.176)
Mother's Age	0.329	(0.364)	25.332	(0.232)
Father's Age	-1.646	(1.217)	36.472	(0.933)
Marital Status	-0.021	(0.023)	1.735	(0.014)
Doctor	-0.011	(0.024)	0.333	(0.015)
Hospital	0.014*	(0.008)	0.959	(0.006)
Number of Children	0.102	(0.087)	1.871	(0.035)

Notes: Column 1 presents the coefficient for the difference between households enrolled and non-enrolled at baseline in a regression of each variable on an indicator for being enrolled at baseline. Column 2 presents the coefficient and standard errors for the non-enrolled group mean. Regressions include the observations for which there is data on baseline enrollment (N=1,612). * p-value<10% ** p-value<5% *** p-value<1%.

TABLE O-7

TREATMENT BALANCE AT THE FAMILY LEVEL

	Full Sample (N=1,832)				Enrolled (N=606)				Non-Enrolled (N=1,006)			
	Difference T-C		Control Mean		Difference T-C		Control Mean		Difference T-C		Control Mean	
	(1)		(2)		(3)		(4)		(5)		(6)	
Panel A: SES characteristics												
Household size	0.114	(0.126)	4.837	(0.101)	0.060	(0.172)	4.852	(0.144)	0.146	(0.158)	4.829	(0.118)
Durable goods	0.069	(0.179)	4.558	(0.160)	0.221	(0.259)	4.694	(0.231)	-0.023	(0.182)	4.477	(0.159)
Owns Dwelling	0.022	(0.039)	0.345	(0.032)	0.116**	(0.058)	0.311	(0.046)	-0.035	(0.042)	0.366	(0.036)
Mother head of hh	0.010	(0.027)	0.826	(0.021)	-0.035	(0.047)	0.857	(0.029)	0.037	(0.031)	0.807	(0.025)
Mother $\leq$ Basic Inc.	-0.022	(0.031)	0.203	(0.026)	-0.017	(0.046)	0.194	(0.038)	-0.025	(0.034)	0.209	(0.029)
Mother $\leq$ Basic	-0.051	(0.039)	0.411	(0.032)	-0.038	(0.057)	0.378	(0.046)	-0.059	(0.040)	0.431	(0.033)
Mother $\leq$ HS	0.029	(0.028)	0.820	(0.022)	0.041	(0.039)	0.816	(0.032)	0.022	(0.032)	0.822	(0.025)
Panel B: Baseline school choice												
Already enrolled	0.002	(0.048)	0.375	(0.038)	-	-	-	-	-	-	-	-
Another child	-0.012	(0.032)	0.416	(0.024)	-0.008	(0.051)	0.420	(0.039)	-0.015	(0.037)	0.414	(0.029)
Panel C: School choice 2011												
Will apply on 2011	-0.015	(0.041)	0.759	(0.033)	0.061	(0.069)	0.776	(0.062)	-0.069	(0.045)	0.743	(0.033)

Notes: Odd columns report the treatment–control difference and its standard error from a regression of each variable on treatment status. Even columns report the control group mean and its standard error. Panels split by reported enrollment status at the preschool meeting; 1,612 survey observations have non-missing enrollment status. Mother's education indicators are cumulative: "$\leq$ Basic Inc." = at most incomplete basic education; "$\leq$ Basic" = at most completed basic education; "$\leq$ HS" = at most completed high school. These are nested indicators, not mutually exclusive categories. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

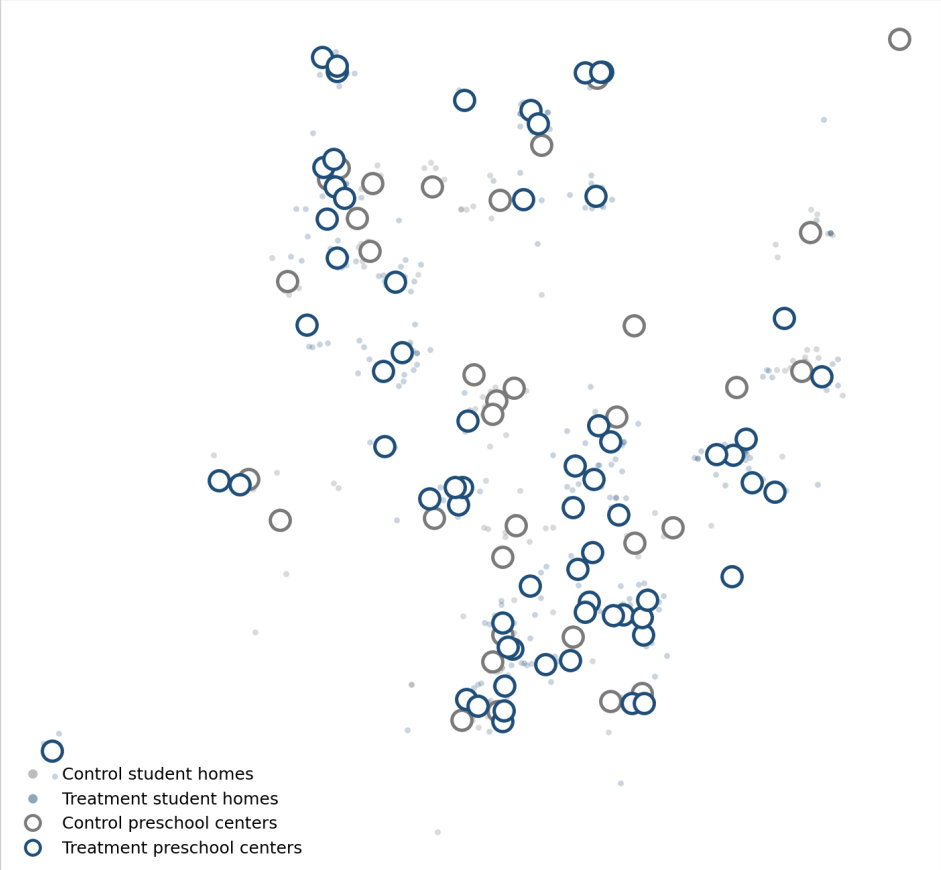
TABLE O-8

SUMMARY STATISTICS FOR ESTIMATION DATA

	Mean	Std. Dev.	10th pctl.	90th pctl.
Panel A: School characteristics				
Quality (value-added, s.d.)	0.097	0.581	−0.653	0.869
First-grade enrollment	46.3	35.4	13	88
Monthly copayment (CLP)	779	1,766	0	3,182
Share priority students	0.441	0.249	0.062	0.750
Public school	0.345			
Private voucher school	0.549			
Private unsubsidized school	0.106			
SEP eligible	0.335			
Panel B: Market structure				
School-year observations		14,378		
Unique schools		3,842		
Markets		53		
Years		4 (2006, 2007, 2011, 2012)		
Residential nodes		3,392		
Schools per market-year	67.8	208.1	9	94

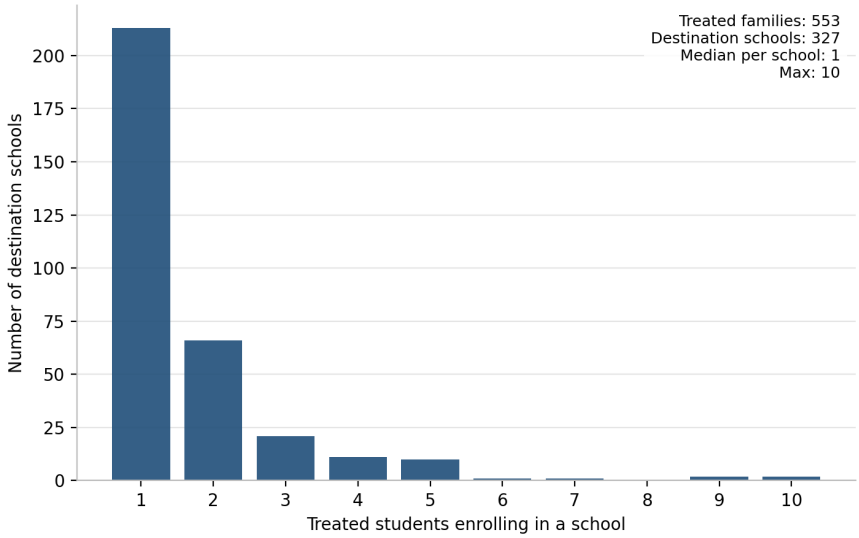
Notes: Panel A reports school-level characteristics for the baseline demand estimation sample (2006, 2007, 2011, 2012). Quality is the school-level value-added measure described in Section 3. Monthly copayment is the out-of-pocket tuition charged by schools; public and most voucher schools charge zero. Panel B summarizes market structure. The supply estimation uses a 10-year panel (2006–2015) with the same 53 markets. Family-level characteristics for the experimental sample are reported in the balance tables above.

FIGURE O-5.—Treatment and Control Preschool Centers and Student Homes in Santiago



Notes: Small dots denote geocoded student homes in the experimental sample. Larger hollow circles denote preschool-center centroids, computed from the homes of children attending each center; colors distinguish treatment and control status. In the 2011 market data, Santiago contains 75,396 first-grade students. The experimental sample in this market includes 670 students, of whom 448 are treated.

FIGURE O-6.—Treated Students per Destination School



O-6. *Reduced-Form Control Robustness*

Tables O-9 and O-10 replicate the main treatment-effect regressions (Tables 1 and 2) with randomization controls but without family controls. The school-choice and fourth-grade achievement estimates for families reached before enrollment are similar to the main specification. Table O-11 then reports the Panel-B PAES estimates across complete blocks of family controls. The PAES estimates are sensitive to adjustment for household characteristics, while the Panel-B eighth-grade mathematics estimate remains positive across specifications.

TABLE O-9

EFFECTS ON CHARACTERISTICS OF SCHOOLS CHOSEN (WITHOUT FAMILY CONTROLS)

	Distance (kms)	Positive Price	Lang 2nd	Lang 4th	Math 4th	V. Added[a]
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Full Sample						
Treatment	0.088	0.049	0.004	0.006	0.014	0.027
	(0.061)	(0.036)	(0.023)	(0.028)	(0.029)	(0.032)
Control mean	1.081	0.494	-0.119	-0.095	-0.082	-0.024
N obs.	1,427	1,701	1,693	1,692	1,692	1,701
Panel B: Already enrolled at the time of the PreK visit						
Treatment	-0.147	-0.001	-0.028	-0.023	-0.044	-0.031
	(0.132)	(0.055)	(0.044)	(0.051)	(0.061)	(0.062)
Control mean	1.192	0.589	-0.031	-0.007	0.046	0.067
N obs.	512	578	576	577	577	578
Panel C: Not enrolled at the time of the PreK visit						
Treatment	0.205***	0.125***	0.051*	0.039	0.078**	0.094**
	(0.069)	(0.041)	(0.028)	(0.031)	(0.034)	(0.038)
Control mean	1.010	0.434	-0.179	-0.137	-0.146	-0.067
N obs.	800	934	928	926	926	934

Notes: Same specification as Table I but without family controls. Standard errors in parentheses are clustered at the preschool-center level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE O-10

EFFECTS ON LONG-TERM STUDENT OUTCOMES (WITHOUT FAMILY CONTROLS)

	SIMCE Test Scores				Reached	College Entrance Exam (PAES)		
	4th Grade		8th Grade			Took	(Cond. on taking PAES)	
	Lang	Math	Lang	Math	12th Grade	PAES	Lang	Math
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Full Sample								
Treatment	0.080	0.108**	-0.003	0.208**	-0.008	0.024	0.176**	0.145***
	(0.057)	(0.054)	(0.081)	(0.102)	(0.035)	(0.035)	(0.072)	(0.056)
Control mean	-0.222	-0.145	-0.059	-0.103	0.728	0.592	-0.367	-0.225
N obs.	1,583	1,584	610	615	1,775	1,775	1,083	1,060
Panel B: Already enrolled at the time of the PreK visit								
Treatment	-0.078	-0.085	-0.190	0.364**	-0.043	0.008	0.242**	0.296***
	(0.117)	(0.097)	(0.228)	(0.171)	(0.054)	(0.060)	(0.110)	(0.094)
Control mean	-0.039	0.058	0.169	0.044	0.770	0.613	-0.287	-0.145
N obs.	543	533	241	243	596	596	373	364
Panel C: Not enrolled at the time of the PreK visit								
Treatment	0.218***	0.202***	0.090	0.207*	0.003	0.027	0.195**	0.150**
	(0.079)	(0.066)	(0.121)	(0.112)	(0.042)	(0.037)	(0.083)	(0.068)
Control mean	-0.339	-0.252	-0.188	-0.162	0.706	0.579	-0.447	-0.284
N obs.	857	865	301	301	975	975	581	568

Notes: Same specification as Table II but without family controls. Standard errors in parentheses are clustered at the preschool-center level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE O-11

PANEL-B PAES ESTIMATES ACROSS FAMILY-CONTROL SPECIFICATIONS

	PAES Language			PAES Mathematics		
Family controls	Coef.	(SE)	p	Coef.	(SE)	p
Maternal education only	0.242	(0.111)	0.031	0.292	(0.096)	0.003
Mother education + child sex	0.204	(0.107)	0.061	0.215	(0.096)	0.028
Mother education + birth-record demographics	0.230	(0.106)	0.031	0.234	(0.098)	0.019
Standard block, excluding SEP eligibility	0.168	(0.109)	0.126	0.160	(0.098)	0.104
Expanded family controls (main)	0.165	(0.109)	0.133	0.159	(0.098)	0.109
Standard block + rich birth records	0.210	(0.105)	0.048	0.183	(0.095)	0.057
Standard block + detailed survey education	0.172	(0.112)	0.130	0.164	(0.103)	0.114
All preceding control blocks	0.208	(0.110)	0.063	0.184	(0.101)	0.071

Notes: Each row reports the treatment coefficient for already-enrolled families using the same randomization controls, preschool-municipality fixed effects, and preschool-clustered standard errors. Birth-record demographics add child sex, maternal marital status and age at birth, twin status, and sibling count. The standard block adds child sex, maternal marital status, household size, durable goods, and home ownership; the expanded block used in the main specification also includes first-grade SEP priority eligibility. Rich birth records additionally include birth weight and length, gestational age, and paternal education when available. The detailed survey-education specification adds the full set of reported education categories. All blocks include corresponding missing-value indicators and are applied as blocks rather than selected separately by outcome.

TABLE O-12

TREATMENT-CONTROL DIFFERENCES IN ADDED FAMILY CONTROLS

	Full sample	Already enrolled	Not yet enrolled
Child is male	0.082***	0.107*	0.082**
	(0.025)	(0.055)	(0.033)
Mother married at birth	0.007	-0.013	0.037
	(0.024)	(0.043)	(0.031)
SEP priority eligible	0.018	0.013	0.013
	(0.024)	(0.049)	(0.033)
Household size	0.121	0.010	0.310**
	(0.140)	(0.211)	(0.148)
Durable-goods index	0.093	0.268	0.207
	(0.146)	(0.228)	(0.176)
Owns dwelling	-0.005	0.127**	-0.040
	(0.030)	(0.063)	(0.028)

Notes: Each entry is the treatment coefficient from a separate regression of the listed family characteristic on treatment, the randomization controls, maternal-education indicators, and preschool-municipality fixed effects. Standard errors in parentheses are clustered by preschool. Missing source values are set to zero and accompanied by missing-value indicators in the outcome regressions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE O-13

COVERAGE OF THE STANDARD EXPANDED FAMILY CONTROLS

	Full sample	Already enrolled	Not yet enrolled
Child sex	98.3%	98.5%	97.9%
SEP priority eligibility	98.3%	98.5%	97.9%
Maternal education	98.1%	98.2%	97.9%
Household size	88.3%	100.0%	100.0%
Durable-goods index	88.3%	100.0%	100.0%
Home ownership	88.2%	100.0%	99.9%
Maternal marital status at birth	99.4%	99.3%	99.6%
Number of children	1,780	596	975

Notes: Entries report the percentage of children with a non-missing value before missing-indicator coding. The 209 children missing household size and durable goods are exactly the 209 children without enrollment-status classification. Consequently, household size and durable goods are observed for every child in Panels B and C; home ownership is missing for one child in Panel C. Missing values in all regressions are set to zero and accompanied by a missing-value indicator, so controls do not reduce the estimation sample.

O-7. *Estimating School Value Added*

Our main measure of school quality is a school-level value-added estimate that captures the school's contribution to student achievement after conditioning on observable student characteristics. Following Neilson (2026), we estimate

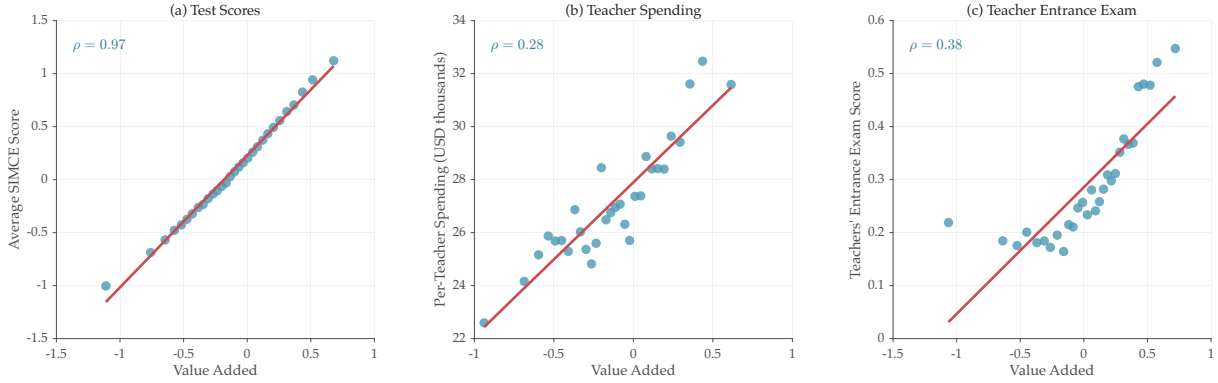
$$Y_{ist} = X'_{ist}\beta + \delta_{jt} + \varepsilon_{ist},$$

where Y_{ist} is the standardized test score of student i at school j in year t, X_{ist} is a vector of student characteristics (including gender, age, SEP eligibility, and birth outcomes from administrative health records), and δ_{jt} is a school-by-year fixed effect. The value-added measure q_j is the average of the estimated $\hat{\delta}_{jt}$ across years, normalized to have mean zero and unit standard deviation across schools. We average over multiple years to reduce noise and obtain a more stable measure of persistent school quality. The controls in X_{ist} do not include lagged test scores, since many students in our sample are entering first grade and have no prior score. Neilson (2026) shows that value-added rankings are very similar whether or not prior scores are included, conditional on the rich set of student background characteristics available in Chilean administrative data.

O-7.1. *Correlations between value added and other measures of school quality*

Figure O-7 provides descriptive evidence on these correlations. Panel (a) plots school-level value-added against raw SIMCE test scores—the quality measure displayed in the report cards given to treated families. The two measures are strongly positively correlated, so the information parents received through the intervention is closely aligned with the value-added measure used in the structural model. Panel (b) shows that schools with higher value-added spend more per teacher on salaries. Panel (c) links value-added to teachers' scores on Chile's national university entrance exam (PSU, now called PAES), providing external validation by connecting a measure of school output to a measure of teacher quality that is entirely independent of student test scores.

FIGURE O-7.—Value Added and Other Measures of School Quality



Notes: Each dot is a bin average (30 bins) computed from school-level means pooled across years. The horizontal axis in all panels plots school-level value-added estimated from the model described in Section O-7. Panel (a): raw SIMCE test scores (average of Math and Reading), the quality measure displayed in the report cards given to treated families. Panel (b): per-teacher expenditure on salaries (USD thousands, trimmed at the 99th percentile). Panel (c): average score on Chile's national university entrance exam (PSU, now PAES) of the school's teachers. Lines show linear fits; ρ denotes the school-level Pearson correlation computed on the underlying (non-binned) data.

O-8. *Additional Supply-Side Results*

TABLE O-14

SUPPLY MODEL ESTIMATES

	Estimate	SE	t-stat
γ_{SEP} — SEP-participation cost shifter	0.212	0.032	6.58
Unit cost of quality (c_j)			
Voucher schools (mean)	0.349		
Private schools (mean)	1.103		
Type-specific time trends ($\delta_{type} \cdot t$)			
δ_{voucher}	0.006	0.004	1.82
δ_{private}	−0.046	0.028	−1.63
Capacity constraint regime			
Constrained obs. (of 22,882)	2,414 (10.5%)		
Ω std (unconstrained)	1.200		
Ω std (constrained)	0.167		

Notes: Two-regime GMM estimates of the quality first-order conditions. Sample: voucher and private non-voucher schools, 2006–2015, 53 markets (22,882 school-year observations, 2,389 schools). Market-year-type effects absorb common cost shocks within each school type and local market. γ_{SEP} is the coefficient on SEP program participation in the additive cost component. School-specific unit costs of quality c_j are concentrated out; means are reported by school type. Type-specific time trends allow unit quality costs to evolve differently across voucher and private schools. The instrument matrix contains the baseline demand instruments and predetermined school characteristics, their interactions with for-profit and nonprofit-voucher indicators, and a Nearby Teacher Quality Index (Table O-15). Constrained observations are post-2008 SEP school-years with enrollment-to-capacity ratio ≥ 0.95. Ω is the regime-specific structural residual.

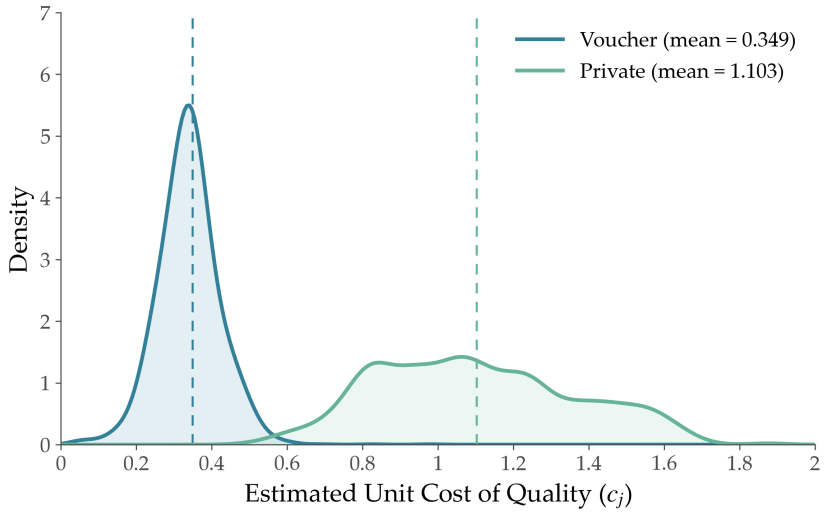
TABLE O-15

INSTRUMENT RELEVANCE FOR SCHOOL QUALITY

	Quality
Exogenous School Characteristics	
Private Non-Voucher	1.435
	(0.974)
For Profit	1.174*
	(0.635)
Religious	0.083
	(0.057)
SEP School	−0.018
	(0.017)
Religious × Voucher	−0.778
	(0.505)
Demand-Side Instruments	
Transfer changes, SEP exposure, local earnings and school characteristics	Yes
Cost-Side Instrument	
Nearby Teacher Quality Index	−0.720***
	(0.124)
Market-Year FE	Yes
R^2 (within)	0.234
F-stat, excluded instruments	56.8
F-stat, Nearby Teacher Quality Index (partial)	32.6
N Obs	22,882
N Market-Years	530

Notes: Regression of observed school quality on the instrument set, included school characteristics, and market-year fixed effects. The excluded demand instruments comprise simulated transfer changes, pre-policy neighborhood SEP exposure and its interaction with the post-reform period, and local non-education earnings deviations. The Nearby Teacher Quality Index is the inverse-distance-weighted average of public-school teacher PSU scores in the same market-year. Heteroskedasticity-robust standard errors are reported. This table documents instrument relevance; the supply GMM moments use the regime-specific first-order-condition residual defined in Section A4. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

FIGURE O-8.—Firm-Level Unit Cost of Quality



Notes: Distribution of the school-specific unit cost of quality c_j concentrated from the quality first-order conditions for voucher and private non-voucher schools, using one estimate per school in the same 2,389-school sample as Table O-14. Means use the full sample. The x-axis is truncated at $c_j = 2$, which covers 99.96% of schools.

O-9. *Unrestricted Demand Estimation*

Table O-16 reports treatment-effect estimates from the unrestricted specification that also estimates the price treatment effect for SEP families ($\psi^{\text{op}}_{\text{SEP}}$). As discussed in Table IV, SEP-eligible families face zero out-of-pocket prices at SEP schools, so this parameter is poorly identified from the experimental variation. The unrestricted estimate is $\hat{\psi}^{\text{op}}_{\text{SEP}} = 0.369$ (SE $= 0.653$), far from significant. All other treatment parameters are qualitatively unchanged: the quality treatment effect for SEP families is 0.476 (SE $= 0.242$) versus 0.500 (SE $= 0.230$) in the restricted specification, and the kappa parameters are similar (1.521 vs. 1.410 for SEP). The GMM objective changes minimally ($Q = 2.637$ unrestricted vs. $Q = 2.922$ restricted), confirming that the restriction is empirically innocuous.

TABLE O-16

UNRESTRICTED DEMAND ESTIMATES (TREATMENT EFFECTS)

	Quality (q)		Price (op)		Distance (d)	
	Coef.	(SE)	Coef.	(SE)	Coef.	(SE)
ψ^c_{SEP}	0.476**	(0.242)	0.369	(0.653)	0.500*	(0.258)
ψ^c_{nonSEP}	0.331	(0.329)	−0.147**	(0.072)	0.100	(0.290)
Scale parameter (H_0: $\kappa = 1$)						
κ_{SEP}			1.521	(0.379)		
κ_{nonSEP}			1.361	(0.564)		
GMM Objective: $Q = 2.637$						

Notes: This table reports treatment-effect parameters from the unrestricted specification that freely estimates $\psi^{\text{op}}_{\text{SEP}}$. Baseline demand parameters (Panel A of Table IV) are unchanged. Standard errors clustered at the preschool-center level. ***, **, *: significant at 1%, 5%, 10%.

O-10. *Counterfactual Simulation and Inference*

The demand-only predictions in Table V involve no assignment simulation. For each perturbation, we re-evaluate choice probabilities and expected school quality. The in-sample prediction uses the unrestricted eight-parameter treatment specification, while the population prediction uses the seven-parameter benchmark imposing $\psi^{op}_{\text{SEP}} = 0$. Numerical delta-method standard errors propagate the preschool-clustered joint covariance of the estimated treatment parameters and are conditional on the baseline demand parameters from Neilson (2026) and the maintained model structure.

The supply equilibrium anchors each private school's competitive-quality term at $\widehat{A}_j = q^0_j + \widetilde{\text{MD}}_j(\mathbf{q}^0, 0)$. We define the demand semi-elasticity and capped markdown as

$$\eta_j(\mathbf{q}, \tau) = (1/s_j)\partial s_j/\partial q_j,$$
$$\widetilde{\text{MD}}_j(\mathbf{q}, \tau) = [\max\{\eta_j(\mathbf{q}, \tau), 1/3.343\}]^{-1}.$$

The same cap applies before and after treatment. Counterfactual quality solves equation 14 jointly across private schools, with public-school quality fixed and each private-school change bounded to ± 0.5 standard deviations. We trace the fixed point by deterministic continuation from no treatment to full treatment. The reporting path uses treatment increments of 0.025; increments of 0.05 and 0.10 produce nearly identical national outcomes. At every step, the

independently evaluated maximum equilibrium residual must not exceed 10^{-6}. Path stability supports the reported aggregate outcome but is not evidence of global uniqueness.

For the capacity-constrained simulations, we generate rank-ordered preferences by drawing $\tilde{U}_{ij} = V_{ij} + \varepsilon_{ij}$ for each family–school pair, with ε_{ij} i.i.d. Gumbel. Each applicant draws one lottery number used at every school; the simulation imposes no SEP or other priority classes. The equilibrium standard errors propagate the covariance of the seven benchmark treatment parameters through the assignment and supply-equilibrium mapping. Each perturbation uses 100 common simulated family draws and assignment priorities per market; population results and standard errors are stable between 50 and 100 draws. These standard errors are conditional on the baseline demand parameters, historical supply estimates, administrative capacities, and the anchored quality FOC. The higher-cost column reports point estimates only.

O-11. *Sensitivity to Peer Composition*

The demand-only counterfactual changes the types of students attending each school while holding school value added fixed. Value added is estimated net of students' own characteristics, but it is measured at the school's observed composition. It can therefore contain the contribution of peers at that composition without describing how measured school quality would change if the student body changed. We use the model-implied change in composition to quantify this omitted channel.

Let z_j^0 and z_j^1 denote the non-SEP share implied by expected enrollment at school j under no treatment and the demand-only full-treatment counterfactual.[4] Let p_{ij}^0 and p_{ij}^1 denote family i's corresponding demand-only choice probabilities. Both are computed with observed value added VA_j, which the demand-only counterfactual holds fixed, and neither is recomputed here. For a given β, we leave baseline expected school quality $\sum_j p_{ij}^0 VA_j$ unchanged and replace its full-treatment counterpart with

$$\sum_j p_{ij}^1 \left[VA_j + \beta(z_j^1 - z_j^0) \right].$$

Thus, the adjusted treatment effect equals the demand-only effect plus $\beta \sum_j p_{ij}^1 (z_j^1 - z_j^0)$, averaged over families. Algebraically, this is equivalent to holding $VA_j - \beta z_j^0$ fixed and evaluating the peer component at the counterfactual composition. Families do not re-sort in response to

[4]The composition measure includes all non-SEP families, including untreated types 5–6.

the adjusted quality and, as in the demand-only benchmark, there are no capacity constraints or school response.

Rather than treating an observational relationship as a causal estimate, we calibrate the size of the composition term. The first two scenarios choose β so that

$$\frac{\mathrm{Var}_w(\beta z_j^0)}{\mathrm{Var}_w(VA_j)} \in \{0.10, 0.20\},$$

where z_j^0 is baseline composition and the variances use baseline expected-enrollment weights. This ratio fixes the scale of the linear composition term. It is not an R^2, an additive decomposition of value-added variance, or an estimate of a causal peer effect. As a stronger cross-sectional benchmark, we also set β equal to the full enrollment-weighted gradient of value added on the non-SEP share. The gradient is 0.23 normalized value-added units for a 10-percentage-point increase in the non-SEP share.

TABLE O-17

DEMAND-ONLY EFFECTS UNDER ALTERNATIVE PEER-COMPOSITION CALIBRATIONS

Peer-composition calibration	Effect per 10 pp.	SEP demand-only effect
No composition adjustment	—	0.19
Composition-term variance ratio: 10%	0.13	0.17
Composition-term variance ratio: 20%	0.18	0.15
Full raw value-added–composition gradient	0.23	0.14

Notes: The first numeric column gives the assumed effect on value added, in the paper's normalized units, of a 10-percentage-point increase in the non-SEP share. The second reports the resulting demand-only treatment effect for SEP families. The 10- and 20-percent rows calibrate the variance ratio defined in the text. The final row uses the full raw value-added–composition gradient and is not interpreted causally. Choice probabilities are the demand-only probabilities computed with observed value added; the composition term enters only the school quality attributed to the full-treatment assignment.

The SEP demand-only effect is 0.19 SD without the composition adjustment. It falls to 0.17 SD under the 10-percent calibration and 0.15 SD under the stronger 20-percent calibration. Assigning the full raw value-added–composition gradient to peers gives an effect of 0.14 SD. Thus, in this accounting exercise, the production channel attenuates but does not eliminate the demand-side gain. These calculations do not address a separate family preference for peer attributes, peer feedback into sorting, or peer composition in schools' objectives.

O-12. *Sensitivity to a Higher Marginal Cost of Quality*

The baseline supply equilibrium preserves the FOC-implied competitive-quality term from the observed equilibrium. If the information policy were implemented at scale and many schools simultaneously tried to improve quality, increased demand for inputs (e.g., skilled teachers) could put upward pressure on the composite marginal cost of quality (Tincani, 2021). Our teacher records do not identify the general-equilibrium wage response to a simultaneous increase in demand across schools. Table O-18 therefore presents an illustrative sensitivity exercise: we multiply private schools' marginal quality costs by 1.05, 1.10, and 1.15, hold the fixed per-student cost terms constant, and re-solve the DA equilibrium with supply response. Operationally, the multiplier rescales the anchored competitive-quality term as shown in equation 15. The no-policy comparison remains the baseline equilibrium at estimated costs.

For SEP families, the equilibrium treatment effect declines from 0.226 SD at estimated costs to 0.184, 0.144, and 0.106 SD as marginal quality costs rise by 5%, 10%, and 15%. The corresponding effects are 0.170, 0.117, 0.067, and 0.020 SD for treated non-SEP families. For all families, including untreated types 5–6, the effects are 0.152, 0.096, 0.044, and -0.005 SD. The 15% case is a qualified point estimate without inference. The exercise therefore shows substantial sensitivity to broad input-cost pressure: the effect remains positive for both treated eligibility groups at 15%, while the all-family average is approximately zero.

Because the marginal cost multiplies the level of the quality index, these scenarios depend on its location normalization; here zero is the 2011 population mean. The exercise also attributes the entire cost increase to implementation at scale and applies it to all private schools, including those whose own demand shifts little. It quantifies sensitivity under that convention rather than estimating the endogenous wage response itself.

Chile's teacher-wage institutions may moderate, but do not eliminate, this response. Municipal-school pay is governed by the national teacher statute, and a national minimum basic salary applies across publicly funded schools. Private voucher schools nevertheless have more flexibility over hiring and supplemental pay and can recruit teachers from municipal or other private schools (Bertoni et al., 2018). The regulated public sector may therefore anchor wages in the broader market, but it does not make teacher supply perfectly elastic or rule out wage pressure when many private schools seek to improve quality at once.

Existing teacher-labor estimates do not provide a direct conversion from these cost changes to wage inflation. For Norwegian teachers, Falch (2010) estimates a school-level labor supply

elasticity of about 1.4, with a range from 1.0 to 1.9, and Falch (2011) reports a short-run elasticity of approximately 1.25 based on teacher mobility. These estimates concern responses to relative wages across schools. Translating them into a system-wide increase in c_{jt} would additionally require the teacher-input share of the marginal cost of quality and a production mapping from teacher inputs to value added, neither of which is identified by our experiment.

TABLE O-18

EQUILIBRIUM TREATMENT EFFECT UNDER A HIGHER MARGINAL COST OF QUALITY

Increase in marginal cost of quality	All families	SEP	Non-SEP
0%	0.152	0.226	0.170
5%	0.096	0.184	0.117
10%	0.044	0.144	0.067
15%	−0.005	0.106	0.020

Notes: Each entry is the equilibrium treatment effect on expected school quality under DA with capacity constraints and supply response. The marginal cost of quality c_{jt} is uniformly scaled upward for all private schools; fixed per-student cost terms are held constant. The quality index uses its demand-scale normalization, with zero at the 2011 population mean. Public schools are held fixed. Entries are point estimates; no standard errors are reported for the imposed cost sensitivities.

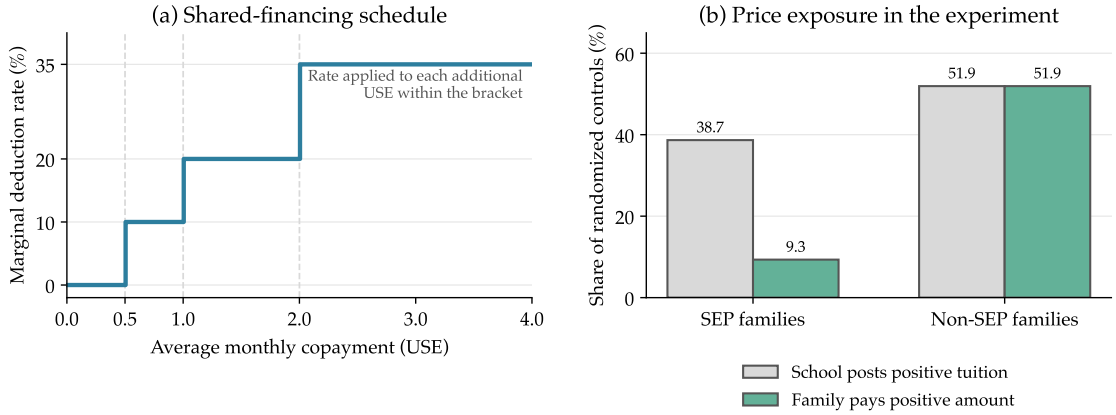
O-13. *Why Tuition Is Held Fixed*

Historical tuition is endogenous in the baseline demand estimation: schools with stronger unobserved appeal may choose different tuition, so the demand model instruments for price when recovering preferences and mean utilities. The counterfactual begins from that estimated historical equilibrium and holds the observed tuition vector fixed. These statements are not contradictory; the first concerns identification of baseline demand, while the second defines the school adjustment margins included in the policy exercise.

The institutional schedule also limits the pricing margin. Under shared financing, schools could charge at most four USE per student per month (DFL No. 2 of 1998, Art. 24). Above 0.5 USE, the state deducted from the regular subsidy 10% of the copayment between 0.5 and 1 USE, 20% of the copayment between 1 and 2 USE, and 35% of the copayment between 2 and 4 USE (Art. 25). The deduction was assessed on the average copayment actually collected, averaged over twelve months and over all enrolled students including those exempted from payment, and between 20% and 100% of it was returned to the school as an earmarked scholarship fund (Arts. 27 and 34), so the effective burden on posted tuition was smaller than the marginal rates suggest. SEP-participating schools were additionally prohibited from charging priority students. Panel (a) of Figure O-9 displays this marginal schedule. Panel (b) shows the resulting

distinction in the structural experimental controls: 38.7% of SEP families chose a school posting positive tuition, but only 9.3% paid a positive amount after applying SEP rules. Among non-SEP controls, both shares are 51.9%. These shares use the 150 SEP and 104 non-SEP controls whose chosen school is matched to the model school set; eight controls are unmatched. The 2015 Inclusion Law subsequently froze existing copayments and began their gradual replacement with public funding. Thus fixed tuition is a natural medium-run benchmark, particularly for SEP families, but it remains a scope restriction rather than a claim that schools could never adjust prices.

FIGURE O-9.—Tuition Regulation and Out-of-Pocket Price Exposure



Priority students pay zero copayment at SEP-participating schools. Panel (b): exact structural-sample controls with observed school outcomes (SEP N=150; non-SEP N=104).

Notes: Panel (a) shows the statutory marginal deduction rate: the share of each additional USE of copayment within a bracket that is deducted from the regular subsidy. Panel (b) reports the share of matched randomized controls whose chosen school posted positive tuition and the share who faced a positive out-of-pocket payment after applying SEP rules. Priority students at SEP-participating schools pay zero copayment. SEP $N = 150$; non-SEP $N = 104$; eight controls have no matched model school.

O-14. *Scale-Up RCT Design Details*

This section supplements Section 7 with the full endpoint-exposure design grid. The calculation is anchored to the solved no-treatment and full-treatment quality equilibria but does not re-solve a partial-treatment equilibrium for each randomized assignment. Instead, baseline demand determines each provider's exposure to treated eligible-family mass, and the linear benchmark maps that exposure into a provider response between the two endpoints. We also report the concave mapping $g(E) = \sqrt{E}$ and the convex mapping $g(E) = E^2$.

O-14.1. *Full Design Space*

The grid varies hexagon radius r, provider-core radius I, and buffer width B while holding within-cluster saturation fixed at $\tau = 0.25$. Geometrically admissible combinations yield 96 candidate designs. For each design, 40 independent assignments determine the centered reference and 200 additional assignments evaluate the design.

The all-assigned target β^* is defined separately for each geometry because changing r, I, or B changes cluster eligibility, the provider sample, and the treated family mass. The first-row panels of the figures report

$$\mathrm{RMSE} = \left\{ \mathbb{E}\left[(\widehat{\beta} - \beta^*)^2 \right] \right\}^{1/2}.$$

The second-row panels report $\mathbb{E}[\widehat{\beta}]/\mathbb{E}[\beta^*]$, the share of the design-specific target recovered by the assignment contrast. Absolute RMSE is therefore not by itself comparable across cells with substantially different targets.

Figure O-10 reports the complete grid under the benchmark linear mapping. For the no-buffer designs displayed in the main text, the highlighted benchmark-conditional choice, $r = 4.0$ and $I = 2.0$ km, has RMSE 0.0046 and recovers 91 percent of its target. Buffers often reduce absolute RMSE within a fixed radius and core, but they also contract the treated region and change the target.

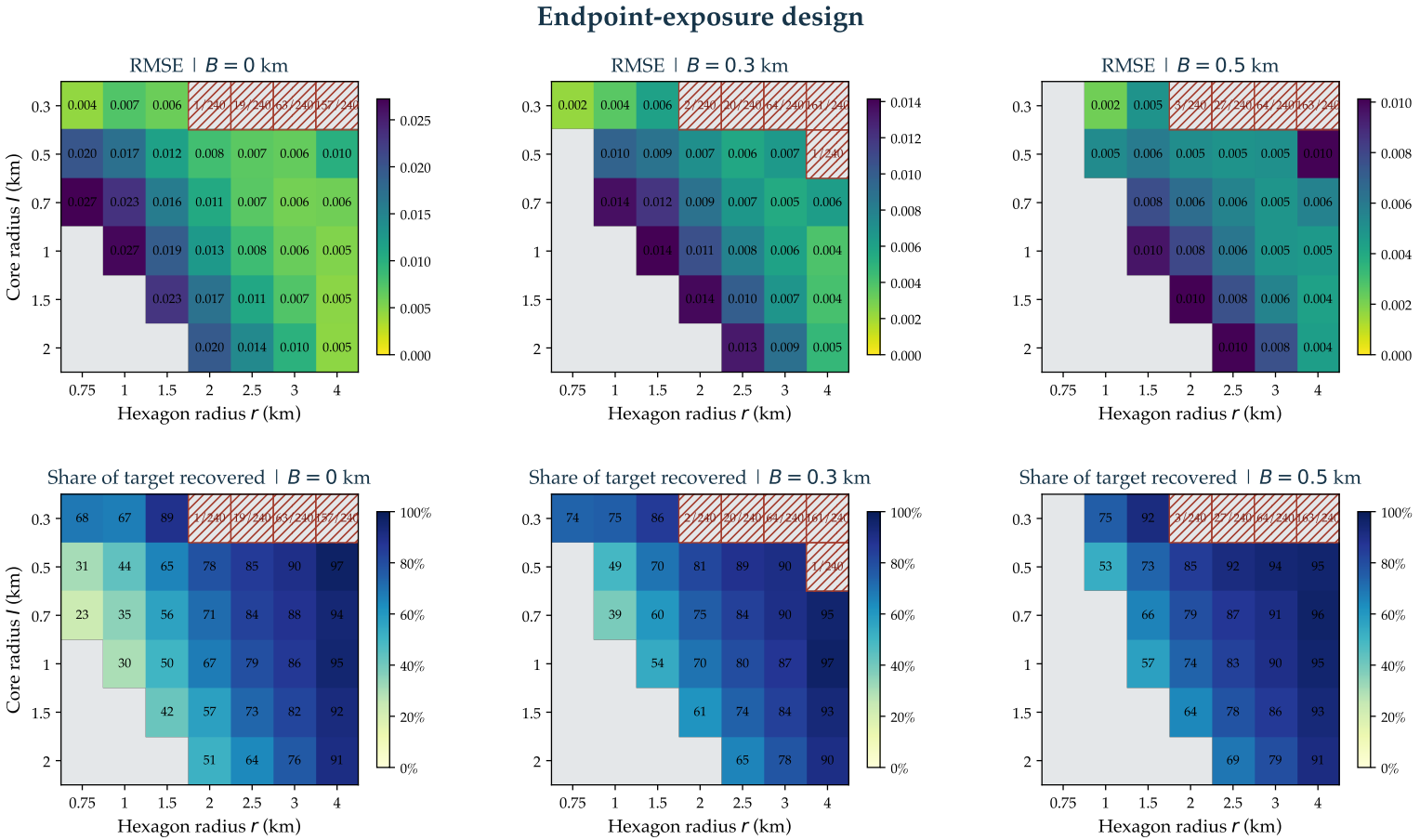


FIGURE O-10.—Endpoint-Exposure Design Grid: Linear Benchmark

Notes: Within-cluster saturation is $\tau = 0.25$. The top row reports RMSE relative to each geometry's all-assigned endpoint-exposure target; the bottom row reports the mean assignment contrast as a share of that target. Each cell uses 40 assignments for the centered reference and 200 evaluation assignments. Gray cells are geometrically infeasible. Hatched labels report the number of assignments, out of 240, for which the provider regression is undefined. No failed assignment is discarded.

O-14.2. *Alternative Exposure Mappings*

Figure O-11 and Figure O-12 repeat the grid under concave and convex mappings between the same solved quality endpoints. Under a no-buffer requirement and a minimum target-recovery rate of 90 percent, all three mappings select $r = 4.0$ km. The preferred core is $I = 2.0$ km under the linear mapping, $I = 0.5$ km under the square-root mapping, and $I = 1.5$ km under the square mapping. The preference for a large hexagon is therefore more stable than the exact core choice.

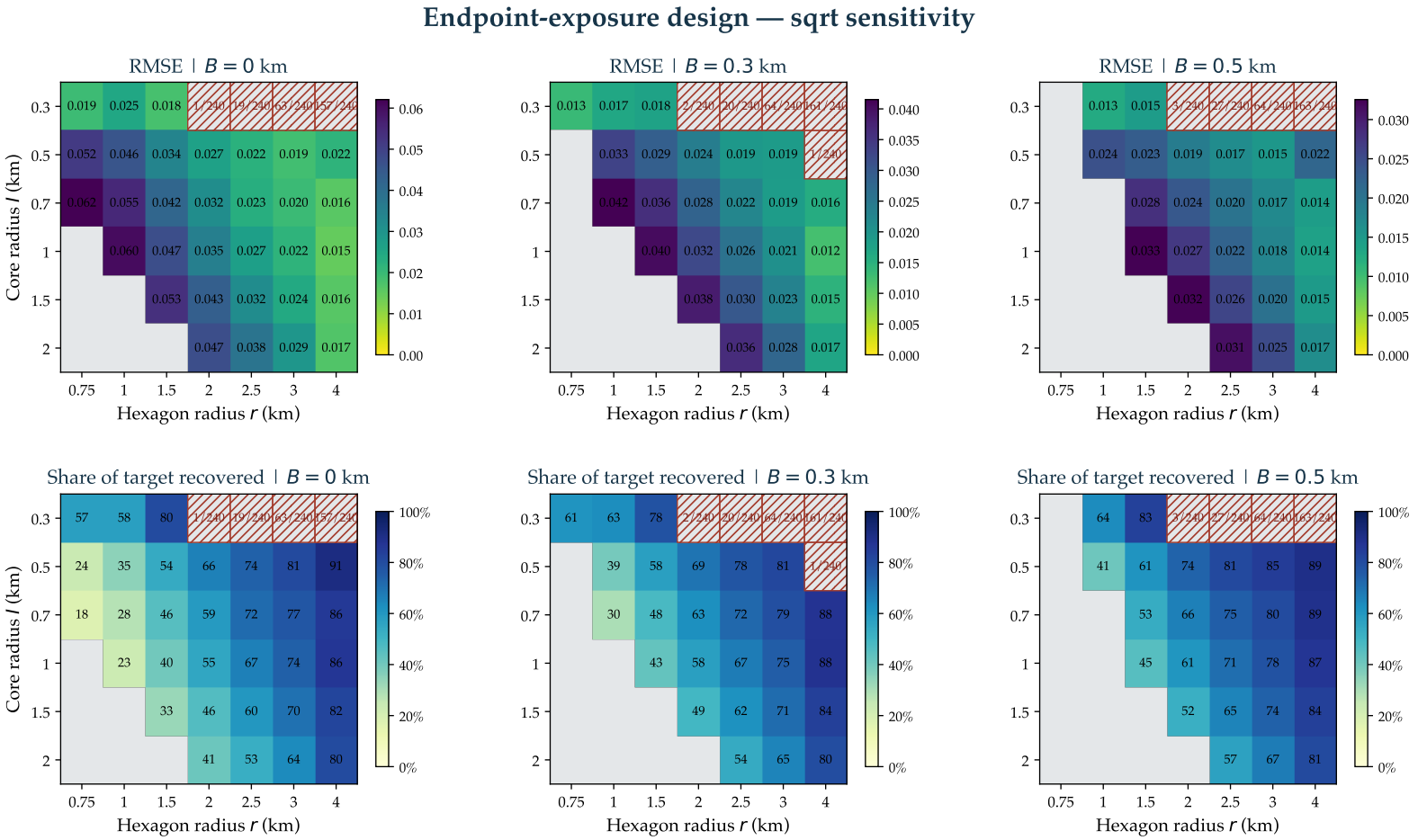


FIGURE O-11.—Endpoint-Exposure Design Grid: Concave Mapping

Notes: This figure uses $g(E) = \sqrt{E}$. All other definitions and simulation counts match Figure O-10.

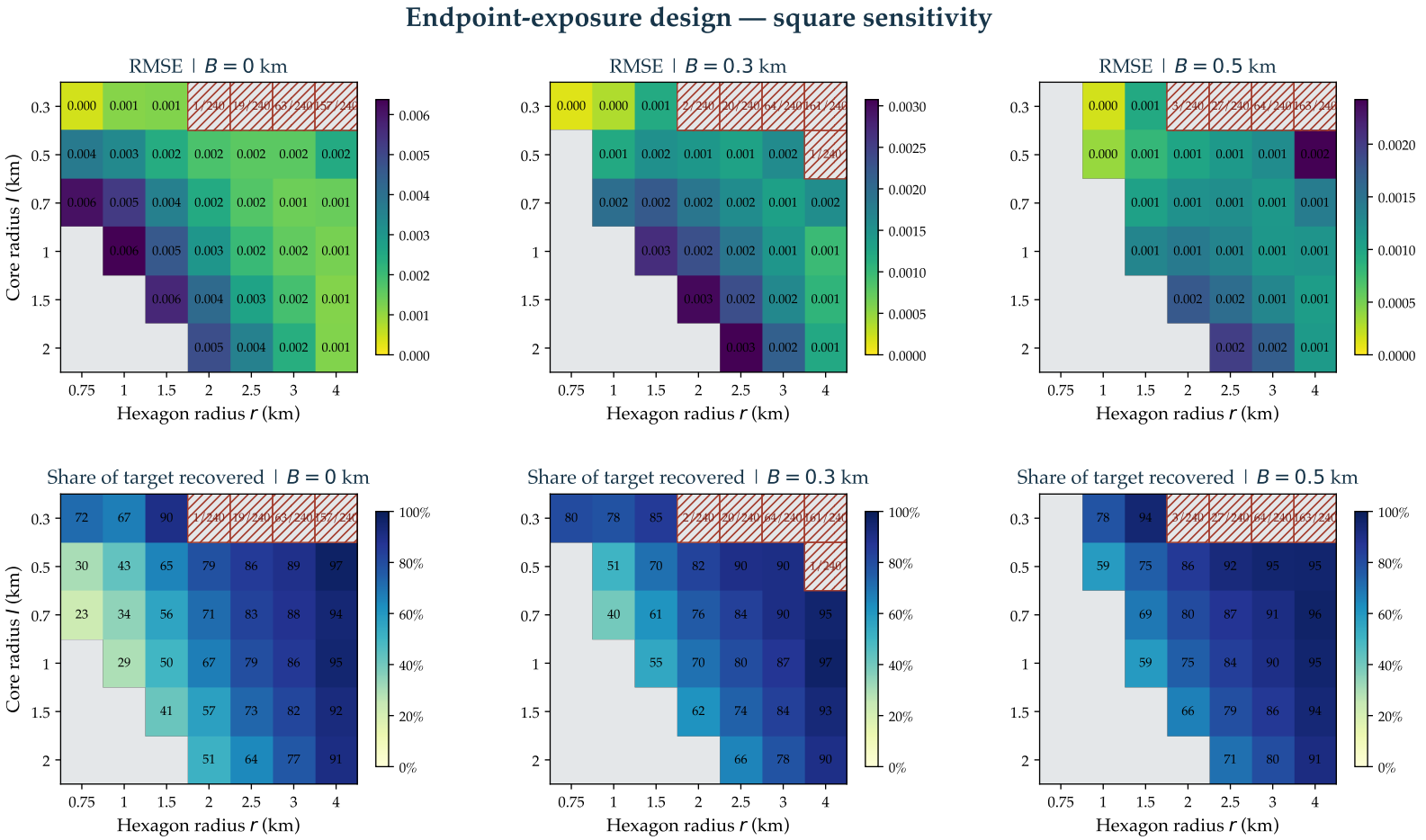


FIGURE O-12.—Endpoint-Exposure Design Grid: Convex Mapping

Notes: This figure uses $g(E) = E^2$. All other definitions and simulation counts match Figure O-10.

Thirteen of the 96 candidate geometries contain at least one assignment for which the provider regression is undefined, generally because a very small core within a large hexagon leaves too few usable treated or control providers. These failures are design outcomes and remain visible in the figures. Conditional rejection of zero is retained in the underlying diagnostics but is not field power: modeled provider quality is observed without measurement error, and the fitted quality endpoints are held fixed.

O-15. *Data Construction*

O-15.1. *Data Usage Across Empirical Exercises*

Because the paper combines an experiment with pre-existing structural estimates and a separate school panel, different empirical objects use different samples and years.

- **Reduced-form experiment.** The information intervention was implemented in the second half of 2010. We measure its effect on school choice using the experimental cohort observed entering primary school in 2011; some children entered their chosen school before first grade.
- **Baseline demand.** We take the first-step demand parameters and recovered mean utilities from Neilson (2026), who estimates the baseline choice model using administrative first-grade choice data from 2006, 2007, 2011, and 2012 across 53 urban markets.
- **Treatment-effect estimation.** The second-step structural estimation uses the subset of experimental families that had not yet enrolled at the time of treatment and that can be matched to the six overlap markets represented in the baseline demand model.
- **Supply estimation.** The supply-side moments are constructed from the 53-market urban school-year panel for 2006–2015 used in the structural estimation. The demand parameters are estimated in 2006, 2007, 2011, and 2012; mean utilities for 2008–2010 and 2013–2015 are recovered by inverting observed market shares at those fixed parameters. Public schools are excluded because their quality investments are governed by municipal budgeting rather than the school-level optimality conditions in the model.
- **Counterfactuals.** The main policy simulations are evaluated in the 2011 market environment, the cohort directly exposed to the intervention.

O-15.2. *School Capacity Imputation*

The capacity-constrained counterfactual in Section 6.1 requires school-level enrollment capacities. Since these are not directly observed in our experimental data, we impute them using multiple administrative sources in the following priority order.

1. **SAE 2020 enrollment slots (public and voucher schools).** We match schools to Chile's 2020 Sistema de Admisión Escolar data by RBD code and use the reported total enrollment slots (*cupos totales*). Because SAE capacities reflect the 2020 physical plant, which may differ from the 2006–2015 estimation period, we adjust for changes in the number of classrooms. Specifically, we infer 2020 classrooms as $\lceil \text{cupos totales}/45 \rceil$ (45 being the legal class-size ceiling) and obtain estimation-period classrooms from administrative enrollment records (distinct first-grade sections per school per year). For schools where these differ, we scale the SAE capacity by the ratio of estimation-period to 2020 classrooms. This adjustment affects roughly 21% of SAE-matched observations. As an additional data-quality floor, observations where enrollment exceeds the adjusted capacity are set to enrollment. This yields matches for 94.5% of public schools and 89.0% of voucher schools in our sample.
2. **Administrative capacity records for private schools.** For private non-voucher schools, we use the number of first-grade classrooms recorded in the 2007–2013 enrollment files, multiplied by the legal class-size ceiling of 45. Private schools without a match are assigned a capacity of 90.
3. **Administrative fallback for unmatched public and voucher schools.** For public and voucher schools not matched to the SAE data, we use the `MaxCap` measure in the same administrative capacity file, which combines the available school capacity and vacancy records.
4. **Market median (last resort).** Any remaining schools receive the median capacity in their market.

A minimum capacity of 5 students is imposed to avoid degenerate markets. The resulting enrollment-to-capacity ratio across all schools has a mean of 0.73 and a median of 0.72, indicating that the imputed capacities produce meaningful but not unrealistic slack. The 10th and 90th percentiles are 0.66 and 0.80, respectively.

REFERENCES

AGUIRRE, JOSEFA (2011): "If you build it they will come: Evidence of the impact of a large expansion of childcare centers over attendance and maternal labor supply," Master's thesis, Pontificia Universidad Catolica de Chile. [1]

BERTONI, ELEONORA, GREGORY ELACQUA, ANALÍA JAIMOVICH, JULIO RODRÍGUEZ, AND HUMBERTO SANTOS (2018): "Teacher Policies, Incentives, and Labor Markets in Chile, Colombia, and Peru: Implications for Equity," Tech. rep., Inter-American Development Bank. [27]

FALCH, TORBERG (2010): "The Elasticity of Labor Supply at the Establishment Level," *Journal of Labor Economics*, 28 (2), 237–266. [27]

——— (2011): "Teacher Mobility Responses to Wage Changes: Evidence from a Quasi-Natural Experiment," *American Economic Review*, 101 (3), 460–465. [28]

JENSEN, ROBERT (2010): "The (Perceived) Returns to Education and the Demand for Schooling," *The Quarterly Journal of Economics*, 125 (2), 515–548. [6]

NEILSON, CHRISTOPHER A. (2026): "Targeted Vouchers, Competition Among Schools, and the Academic Achievement of Poor Students," *Econometrica*, conditionally Accepted. [19, 24, 35]

NGUYEN, TRANG (2008): "Information, Role Models and Perceived Returns to Education: Experimental Evidence from Madagascar," Mimeo, MIT. [6]

OECD (2016): *Starting strong IV: Monitoring quality in early childhood education and care - Chile*, OECD Publishing. [1]

TINCANI, MICHELA M. (2021): "Teacher Labor Markets, School Vouchers and Student Cognitive Achievement: Evidence from Chile," *Quantitative Economics*, 12 (1), 173–216. [27]